UChicago Medicine.

Appointment No-Show & Cancellation

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Business Need & Value

Business Need: Lower the University of Chicago Hospital's percentage of patients not showing up to scheduled appointments

A no-show leads to:

- Wasted resources (staff & doctor's time, consultation room cleaning resources)
- Mismanaged schedule and financial discrepancies

Business Value:

- Improve physician productivity
- Efficient use of hospital resources
- Increase ambulatory (outpatient service) revenue

Background Information

- Current average no show rate is 20%, meaning 1 in 5 appointments are no-shows.
- Groups that are most likely to no-show or cancel:
  - Young adults
  - Pediatric patients: children rely on their parents to take them to appointments
- Higher rate of no shows in returning patients.

Overview Of Our Proposal

- Text/call reminders will be followed up by an additional message after 24 hours if no response is given, where the appointment will simply be **canceled if no response is received 24 hours prior to the scheduled appointment time.**
  - With this time, employees can see if a waitlisted patient would like to fill the open time slot.
  - Upon cancellation, rescheduling options will be sent out to patient.
- Patients will be entered into a **raffle** to award and incentivize good attendance.
- A **survey** will be sent out to patients to collect data on their reasons for a no-show, which will also provide resources to combat no-shows.
- Recommendation to hire an **additional staff member**: more details in later slides

Use Case 1: Current Appointment Reminders and Cancellation

- **ID**: 0001
- **Priority**: High - Addressing No-Shows
- **Actors**: Patient, Hospital Staff
- **Description**: This use case outlines procedures for contacting patients before scheduled appointments to confirm attendance, as well as the MyChart cancellation process
- **Trigger**:
  Upcoming appointment (approaching within 7 days).
- **Type**: Temporal
- **Preconditions**:
  - Patient appointment is in the hospital's scheduling system.
  - Contact details (email, phone number) for the patient are available in the system.

Current Process

1.System Check: Automated check identifies all appointments within the next 7 days.

2.Retrieve Contact Information: System or staff retrieves patient's contact info.

3.Send Reminder Notification: A reminder is sent via email, SMS, or automated phone call. EPIC's patient data can be used to set the frequency of reminders.

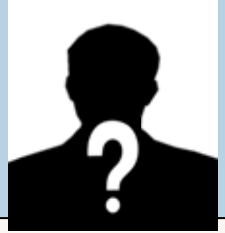
4.Confirmation: The reminders sent via MyChart or text often include options to confirm, reschedule, or cancel the appointment with just a few clicks.

4.a. If patient chooses to cancel, they are asked to provide a reason for cancellation, however, this step is voluntary.

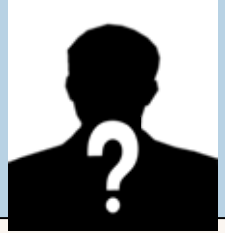
5. Second Reminder (No Response): If no response, follow-up reminders are automatically sent, usually a day or two before the appointment date. Appointment is not cancelled, even if no response is given

Our Proposal: Appointment Cancellation

- Make it mandatory for patients to provide a reason for cancellation.
  - This will give the hospital feedback in the form of data about why patients are not coming in for appointments.
  - Can be used to assess potential issues
  - Simple process:
    - adds only one extra step to the cancellation process.



Text/Call Reminders



- We have decided to leave this as an un-automated process.

- With a current no-show rate of 20%, 1 in 5 appointments being missed is a huge financial loss for the hospital.
  - With the profit margin of decreasing the no-show rate to 10% leaves the hospital with the resources to hire an additional front-desk worker.

- This would be on a trial basis to see how much the no-show rate decreases with these updates.
- This is a process that could be automated in the future, but $22/hr salary is much cheaper than spending thousands on a system that may not be effective.

- The new front-desk worker job responsibilities are:
  - Sending out texts or calls based on patient preferences, with a follow up message after 24 hours if not given a response.
  - Monitor the survey responses and run the mail merge extension
  - Reviewing patient attendance history in patient profile to enter patients into the raffle
  - Announcing the monthly raffle dates, and prizes.

Use Case 2: Missing an Appointment & the Current Follow-Up Process

ID: 0002

Priority: High - Addressing No-Shows

Actor: Hospital Staff

Description: This use case describes the process of patient outreach to those miss their appointments without canceling.

Trigger: Appointment No-Show | **Type**: External

Preconditions: Lack of patient Check-In or Cancellation

Current Process:

1. If a patient misses an appointment, they are contacted by the hospital's appointment scheduling team.
   a. Via phone, email, or MyChart notifications.
   b. Meant to remind the patient of their missed appointment and encourage them to reschedule.
2. During the follow-up, staff may inquire about the reason for the missed appointment, **but a response is not required**
3. Any reasons for cancellation provided by patients are recorded in MyChart patient profile

Our Proposal: Create & Distribute Survey

Potential Issue: If patients don't show up to appointments, they are also unlikely to fill out the survey

Potential Solutions:

- Send the survey to all UChicago Hospital patients, no only those that did not show up to their appointment
- Work with UChicago Laboratory Schools (Middle & High School) teachers to have their students fill out the survey.
  - Similar demographic
  - Use student responses as a representative sample

- Use data gathered in the survey for future communication with patients.
- Based on patient responses, send personalized information and resources.

Survey & Spreadsheet

Survey:
https://docs.google.com/forms/d/1pJyHo4KvusXb_rNxqflGxAxBjrvNzBHT4RHsA7SsgVA/edit

Spreadsheet:

https://docs.google.com/spreadsheets/d/1F1VwmP1eyeN1x0ebljlni4-3VIO-9AEv1lSTXusZlrs/edit?usp=sharing

Email Response:
https://drive.google.com/file/d/18m1UyqnD_9f50QJKrMi__ShliZ9l7xgo/view?usp=sharing

Automated Email Response

UChicago Hospital <katarineoliynyk@gmail.com> Nov 13, 2024, 2:08 PM (2 days ago)

to me

Hi Katherine Oliynyk!

Thank you for completing our survey! Based on your response, we'd like to share the following resource(s) with you:

Public Transportation Options: https://www.uchicagomedicine.org/patients-visitors/patient-information/directions-and-maps/public-transportation
Video Appointment Options: https://www.uchicagomedicine.org/patients-visitors/patient-information/telemedicine/video-visit
:

Follow-Up Care Info: https://drive.google.com/file/d/1IlyL4FA-7oDGPk14M95uGN3_ivC5U6IW/view?usp=sharing
Benefits of Consistent Doctor's Visits: https://drive.google.com/file/d/1j44MuzLg0VqDReOQu5oZm6HKq9FpmsmW/view?usp=sharing

Let us know if there's anything else we can do to assist you!

UChicago Medicine

Yet Another Mail Merge

Free	Personal	Professional
@gmail.com or Workspace account	@gmail.com account	Google Workspace account
$0	$3/month $36 billed annually	$5/month $60 billed annually
All basic features included	**All** basic features included	**All** basic features included
Open, **click** & **response** tracking	**Open**, **click** & **response** tracking	**Open**, **click** & **response** tracking
Unsubscribe & **bounce** tracking	**Unsubscribe** & **bounce** tracking	**Unsubscribe** & **bounce** tracking
~~**Scheduling**~~	**Scheduling**	**Scheduling**
No branded footer	**No** branded footer	**No** branded footer
Up to 20 recipients / day	**Up to 400** recipients / day ⓘ	**Up to 1,500** recipients / day ⓘ
1 user	**1** user	**1** user
Best effort support	**Priority** support	**Priority** support

Use Case 3: Gather Patient Attendance History in EPIC

ID: 0003

Priority: High - Addressing No-Shows

Actor: Hospital Staff

Description: use case describes the current process of gathering patient attendance history using EPIC

Trigger: Staff needs to know patient attendance history

Type: Temporal

Preconditions: Patient data is present in EPIC

Current Process:

1. When an appointment is scheduled (by staff or patients), details are entered into EPIC scheduling system (EpicCare)
2. System generates a unique appointment ID and stores appointment in patient schedule
3. When the patient arrives for their appointment, they check in with the front desk, their identity is verified, and the visit is marked “arrived” / “checked-in”
4. Attendance status is assigned: “Completed”, “No Show”, “Canceled”. Status is updated by front desk staff
5. If patient reschedules, new appointment is generated in the system
6. Attendance history is stored as part of patient data
7. No-show data can be tracked by department, period of time, and patient

Attendance Incentive: Raffle

- Use EPIC Patient Attendance history to enter patients into a raffle
- Create monthly posters and graphics to promote the raffle

Appointment Attendance Raffle

Rules

- Attend 5 appointments without rescheduling or cancelling
- Limited to 3 winners per month

Dates

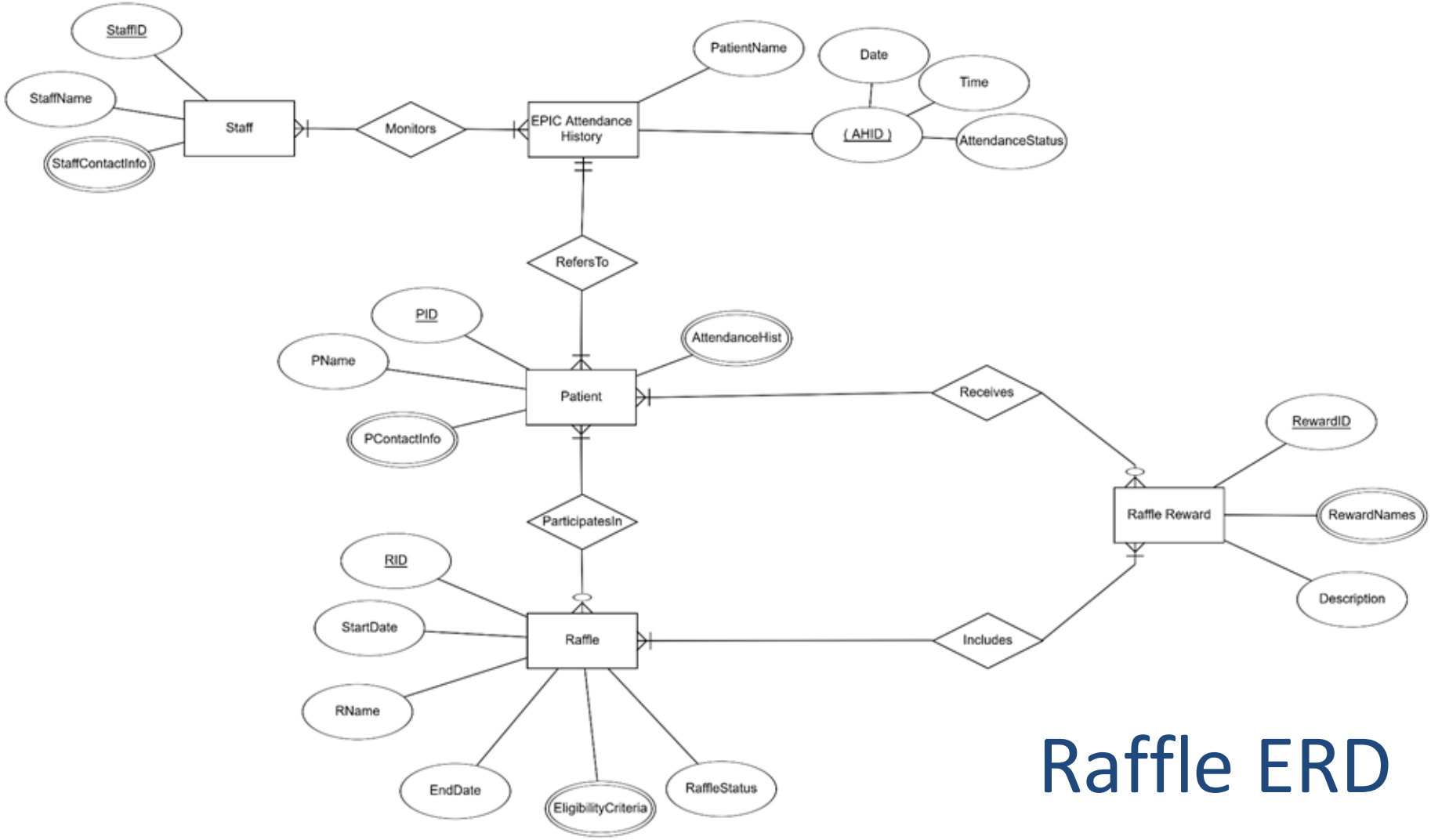
- Raffle will take place the last week of each month
- Patients will be notified if they win by the end of the month

Raffle Items & Rewards (Teens & Adults)

- Gift cards to:
  - Amazon, Starbucks, Dunkin', Apple, etc.

Raffle Items & Rewards (Children & Teens)

- Gift cards to: Steam, Lego, American Girl, etc.
- Toys chosen by survey


StaffID
StaffName
StaffContactInfo
Staff
Monitors
EPIC Attendance History
PatientName
Date
Time
(AHID)
AttendanceStatus
RefersTo
PID
PName
PContactInfo
Patient
AttendanceHist
Receives
RewardID
Raffle Reward
RewardNames
Description
ParticipatesIn
RID
StartDate
RName
Raffle
Includes
EndDate
EligibilityCriteria
RaffleStatus

Raffle ERD

Lost Revenue From No-Shows

- Physician Salary
- Profit loss from inability to book other patients
- Inefficient resource utilization

Raffle Costs

- Low cost: only a set amount of winners.
- Raffle prizes are affordable
- Raffle may make patients more likely to attend appointments, decreasing revenue lost from no-shows

Functional Requirements

(Survey) Data Collection & Storage

01

- Survey responses should be automatically entered into a linked google spreadsheet.
- Survey responses should be timestamped.

Email Response Automation

02

- Survey Responses should trigger customized email responses.
- Email responses can be personalized based on survey responses.
- Automate sending email via mail merge extension.

Analytics

03

- Set appointment reminder frequency based on patient cancellation/no-show history from EPIC.
- Use EPIC attendance history data to enter patients into raffle.
- Require patients give their reason for cancellation in MyChart

Non-Functional Requirements

Operational
01

- Web-based system run on any browser
- Connect to printers wirelessly
- Must integrate with existing hospital systems

Performance
02

- Quick response time
- New customer information is updated routinely
- Scalability & ability to handle a large number of users

Accessibility
03

- System should support multiple devices
- Multilingual support for automated reminders
- Available 24/7

Security
04

- Follows HIPAA guidelines
- Data encryption, backup & recovery
- Manage and record which staff has access to patient information.
- Maintain logs of user interactions, data modifications

Next Steps

How to Implement These Proposals:

- Ownership of Google Form and Spreadsheet for survey will be transferred to hospital staff (or anyone assigned to the role)
- Raffle Advertisement will be available to send to patients and display at hospital
- Implement changes to current appointment reminder and cancellation systems

Thank You!